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United States Patent	12408838
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Tang; Cha-Min et al.

Apparatus and method for tooth pulp vitality detection

Abstract

An apparatus provided for detecting tooth pulp vitality includes a handle, a pivot and a caliper. A pair of fiber optic lines pass through an interior of the handle, the pivot and the caliper. One of the fiber optic lines is a source line with a connector at an end of the source line and another of the fiber optic lines is a detector line with a connector at an end of the detector line. The source line is single mode in the handle and the detector line is multi mode. A system is also provided for detecting tooth pulp vitality that includes the apparatus. Additionally, a method is provided for detecting tooth pulp vitality that employs the apparatus.

Inventors: Tang; Cha-Min (Wayne, PA), Schmitt; Joseph (Cupertino, CA)

Applicant: UNIVERSITY OF MARYLAND, BALTIMORE (Baltimore, MD); THE UNITED STATES GOVERNMENT AS REPRESENTED BY THE DEPARTMENT OF VETERANS AFFAIRS (Washington, DC)

Family ID: 63252968

Assignee: UNIVERSITY OF MARYLAND, BALTIMORE (Baltimore, MD)

Appl. No.: 18/304711

Filed: April 21, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230263398 A1	Aug. 24, 2023

Related U.S. Application Data

continuation parent-doc US 16487775 US 11666223 WO PCT/US2018/019164 20180222 child-doc US 18304711

Publication Classification

Int. Cl.: **A61B5/00** (20060101); **A61B5/026** (20060101); **A61B90/00** (20160101); A61C3/10 (20060101); A61C19/04 (20060101)

U.S. Cl.:

CPC **A61B5/0088** (20130101); **A61B5/0066** (20130101); **A61B5/0261** (20130101); **A61B5/4547** (20130101); **A61B5/682** (20130101); A61B90/361 (20160201); A61B2090/3614 (20160201); A61C3/10 (20130101); A61C19/04 (20130101)

Field of Classification Search

CPC: A61B (5/0088); A61B (5/0066); A61B (5/0261); A61B (5/4547); A61B (5/682); A61B (5/6887); A61B (2090/3614); A61B (2505/05); A61C (3/10); A61C (19/04)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is continuation of U.S. patent application Ser. No. 16/487,775, filed Aug. 21, 2019, which is a 371 National Stage Application of PCT Application No. PCT/US18/19164, filed Feb. 22, 2018, and claims benefit of Provisional Application No. 62/461,955, filed Feb. 22, 2017, the entire contents of which are hereby incorporated by reference as if fully set forth herein, under 35 U.S.C. § 119(e). This application also claims benefit of Provisional Application No. 62/582,518, filed Nov. 7, 2017, the entire contents of which are hereby incorporated by reference as if fully set forth herein, under 35 U.S.C. § 119(e).

BACKGROUND

(1) Motion sensors have a wide range of applications and are implemented using a wide range of techniques exploiting a wide range of physical principles. Medical applications for motion sensors include diagnosis and treatment of a subject based on the presence, function or location of blood vessels that carry moving blood cells and other constituents. Sometimes the tissue, such as bone, is difficult to penetrate for obtaining the motion measurement. Only a few motion sensors are suitable for such varied medical applications.

(2) One technique that has been used in medical applications is laser Doppler flowmetry (LDF). As stated in Marc F. Swiontkowski, "Laser Doppler Flowmetry—Development and Clinical Application," *Iowa Orthopaedic Journal*, v11, pp119-126 (1991), "Laser Doppler Flowmetry (LDF) is an accurate and reliable method for assessing microcirculatory function. Through a series of in vitro and in vivo experiments, LDF output has been shown to be reproducible and to correlate with bone blood flow as estimated by other methods. The utility of the method in assessing meniscal, tendonous, and ligamentous perfusion has also been demonstrated. LDF has proven potential in clinical research in osteonecrosis, osteomyelitis, fracture healing, and other areas." LDF has also been applied to neurosurgery, dermatology and dentistry. However, these devices suffer from low signal to noise ratio and low spatial resolution compared to blood vessels of interest during many diagnosis and treatment procedures.

(3) Another technique used in medical applications is Doppler optical coherence tomography (DOCT). Optical coherence tomography (OCT) is an optical signal acquisition and processing method. It captures micrometer-resolution, three-dimensional images from within optical scattering media (e.g., biological tissue). Depending on the properties of the light source (superluminescent diodes, ultrashort pulsed lasers and supercontinuum lasers have been employed), optical coherence tomography has achieved sub-micrometer resolution (with very wide-spectrum sources emitting over a range of wavelengths about 100 nanometers wide, 1 nanometer, nm, =10^{sup.}-9 meters). Commercially available optical coherence tomography systems are employed in diverse applications, including diagnostic medicine, notably in ophthalmology where it can be used to obtain detailed images from within the retina. Recently it has also begun to be used in interventional cardiology to help diagnose coronary artery disease. Determining the Doppler shift in the returned signal due to motion of the optical scatterers in the sample, turns the OCT into a 3D imaging DOCT motion sensor. However, these devices are expensive, complicated, unwieldy and difficult to use when diagnosing or treating patients.

(4) Yet another technique used in medical applications is Coherence Gated Doppler (CGD) Motion Sensing. CGD involves adjusting a bandwidth and corresponding coherence length of an optical

source for purposes of capturing Doppler shifted signals from a target volume to measure motion within the target volume.

(5) Testing the vitality of the tooth is a common procedure in dentistry when evaluating the need for a root canal procedure as a consequence of tooth decay, infection or trauma. Most available testing methods rely on subjective assessment of the sensitivity of the tooth to cold, heat, or electrical stimulation. Many dentists use a simple pulp vitality test based on application of ice or heated rubber to the tooth. To improve repeatability, others use specialized instruments that generate a localized source of heat or cold that can be applied to the tooth at a controlled temperature. Since the first invention disclosures in the early 1940's, numerous methods for electrical stimulation of the tooth have been developed, which have led to the variety of electrical pulp stimulators available in the market today. These existing methods have a sensitivity and specificity in the 80% to 90% range.

SUMMARY

(6) Techniques are provided for simple, inexpensive, reliable, easily-used motion sensors with sufficiently high sensitivity to detect tooth pulp flow and sufficiently high specificity to distinguish it from the blood flow arising from the surrounding gum tissue and the periodontal ligament. It is here recognized that current LDF techniques are inadequate for reliable tooth pulp vitality detection. In one example, current LDF techniques feature inadequate sensitivity to pulp flow, particularly in large molars. Specifically, current LDF techniques do not account for variance in tooth thickness when measuring pulp flow. In another example, current LDF techniques are susceptible to blood flow arising outside of the tooth. Specifically, current LDF techniques are unable to isolate scattered signals from the pulp region while excluding scattered signals from tissue that immediately surrounds the tooth (e.g. gum tissue and periodontal ligament). In yet another example, current LDF techniques do not permit the use of low-cost, reusable probes that are particularly advantageous when performing tooth pulp vitality detection in multiple individuals. Specifically, current LDF techniques employ single mode fibers which require complex single mode to single mode coupling for a detachable and reusable probe tip. The system and method for tooth pulp vitality detection described herein overcomes one or more of these noted drawbacks of current LDF techniques.

(7) In a first set of embodiments, an apparatus for detecting tooth pulp vitality includes a handle, a pivot and a caliper. A pair of fiber optic lines pass through an interior of the handle, the pivot and the caliper. One of the fiber optic lines is a source line with a connector at an end of the source line and another of the fiber optic lines is a detector line with a connector at an end of the detector line. The source line is single mode in the handle and the detector line is multi mode.

(8) In a second set of embodiments, a system for detecting tooth pulp vitality includes a probe with a tip configured to engage opposite sides of a subject. The probe includes a pair of fiber optic lines passing through an interior of the probe. The fiber optic lines include a source line that is predominantly single mode and includes a connector at one end and a detector line that is multi mode and includes a connector at one end. The system also includes an optical source to emit output light with an adjustable bandwidth, where the optical source is coupled to the source line connector to transmit the output light into the source line. The system also includes a reference path with an adjustable delay line coupled to the optical source to receive the output light from the optical source. The system also includes a detector coupled to the adjustable delay line to receive the output light from the reference path and also coupled to the detector line connector to receive scattered light. The system also includes a processor and a memory with one or more sequence of instructions. The memory and the sequence of instructions, with the processor, are configured to cause the system to adjust the adjustable delay line of the reference path such that a zero delay plane is positioned in a target volume of the subject, where the zero delay plane is defined such that an optical path of output light through the reference path to the detector is equal to an optical path of scattered light from the zero delay plane through the detector line to the detector. The memory

and sequence of instructions, along with the processor, are further configured to cause the system to adjust the bandwidth of the output light to vary a coherence length of the output light such that scattered light from the target volume and output light through the reference path generate an interference signal at the detector and such that scattered light from outside the target volume and output light through the reference path do not generate an interference signal at the detector. The memory and sequence of instructions, along with the processor, are further configured to cause the system to determine motion within the target volume based on a Doppler shift between scattered light from the target volume and the output light through the reference path, wherein the Doppler shift is based on the generated interference signal.

(9) In a third set of embodiments, a method for detecting tooth pulp vitality includes engaging opposite sides of a tooth of a subject with a tip of a probe, where the probe includes a pair of fiber optic lines passing through an interior of the probe. The fiber optic lines include a source line that is predominantly single mode and includes a connector at one end and a detector line that is multi mode and includes a connector at one end. The method further includes emitting output light from an optical source with an adjustable bandwidth, where a first portion of the output light is coupled into the source line connector and a second portion of the output light is coupled into a reference path with an adjustable delay line. The method further includes receiving, at a detector, output light from the reference path and scattered light from the detector line connector. The method further includes adjusting the delay line of the reference path with a processor such that a zero delay plane is positioned in the pulp region of the tooth, where the zero delay plane is defined such that an optical path of output light through the reference path to the detector is equal to an optical path of scattered light from the zero delay plane through the detector line to the detector. The method further includes adjusting the bandwidth of the output light with the processor to vary a coherence length of the output light such that scattered light from the pulp region and output light through the reference path generate an interference signal at the detector and such that scattered light from gum tissue and output light through the reference path do not generate an interference signal at the detector. The method further includes determining motion within the pulp region with the processor based on a Doppler shift between scattered light from the pulp region and output light through the reference path, where the Doppler shift is based on the generated interference signal.

(10) Still other aspects, features, and advantages are readily apparent from the following detailed description, simply by illustrating a number of particular embodiments and implementations, including the best mode contemplated for carrying out the invention. Other embodiments are also capable of other and different features and advantages, and its several details can be modified in various obvious respects, all without departing from the spirit and scope of the invention. Accordingly, the drawings and description are to be regarded as illustrative in nature, and not as restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments are illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings in which like reference numerals refer to similar elements and in which:

(2) FIG. 1A through FIG. 1C are block diagrams that illustrate an example method to produce an interference signal at a detector, which indicates motion in a target volume;

(3) FIG. 2A is a block diagram that illustrates an example laser Doppler flowmetry (LDF) apparatus for detecting fluid flow within tissue;

(4) FIG. 2B is a block diagram that illustrates an example Doppler optical coherence tomography (DOCT) apparatus for producing images of fluid flow location within a tissue;

- (5) FIG. 2C is a block diagram that illustrates an example laser Doppler flowmetry (LDF) apparatus for detecting fluid flow within tooth pulp;
- (6) FIG. 3A is a block diagram that illustrates example components of a coherence gated Doppler (CGD) system, according to various embodiments;
- (7) FIG. 3B is a block diagram that illustrates example components of an apparatus used in the system of FIG. 3A, according to various embodiments;
- (8) FIG. 3C is a block diagram that illustrates example components of an apparatus used in the system of FIG. 3A, according to various embodiments;
- (9) FIG. 3D is a block diagram that illustrates example source line of the apparatus of FIG. 3C, according to various embodiments;
- (10) FIG. 4 is a flowchart that illustrates an example of a method for detecting tooth pulp vitality, according to various embodiments;
- (11) FIG. 5A is a block diagram that illustrates an example of positioning of a tip of the probe on opposite sides of a tooth in the system of FIG. 3A, according to various embodiments;
- (12) FIG. 5B is a plot that illustrates an example of traces indicating an optical path difference between pulp region and gum tissue for various spacing between the probe and gum tissue, according to various embodiments;
- (13) FIG. 6 is a block diagram that illustrates example components of a coherence gated Doppler (CGD) system for detecting tooth pulp vitality, according to various embodiments;
- (14) FIG. 7 is a block diagram that illustrates example components of a coherence gated Doppler (CGD) system for detecting tooth pulp vitality, according to various embodiments;
- (15) FIG. 8 is a block diagram that illustrates an example of a light source used in the system of FIG. 7, according to various embodiments;
- (16) FIG. 9A is a block diagram that illustrates an example of positioning of a tip of the probe on opposite sides of a tooth in the system of FIG. 7, according to various embodiments;
- (17) FIG. 9B is a block diagram that illustrates an example of positioning of a tip of the probe on opposite sides of a tooth below the gum line in the system of FIG. 7, according to various embodiments;
- (18) FIG. 9C is a block diagram that illustrates an example of a tip of the probe of FIG. 3A, according to various embodiments;
- (19) FIG. 10 is a block diagram that illustrates a computer system upon which an embodiment of the invention may be implemented; and
- (20) FIG. 11 is a block diagram that illustrates a chip set upon which an embodiment of the invention may be implemented.

DETAILED DESCRIPTION

- (21) A method and apparatus are described for tooth pulp vitality detection. In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the present invention. It will be apparent, however, to one skilled in the art that the present invention may be practiced without these specific details. In other instances, well-known structures and devices are shown in block diagram form in order to avoid unnecessarily obscuring the present invention.
- (22) Notwithstanding that the numerical ranges and parameters setting forth the broad scope are approximations, the numerical values set forth in specific non-limiting examples are reported as precisely as possible. Any numerical value, however, inherently contains certain errors necessarily resulting from the standard deviation found in their respective testing measurements at the time of this writing. Furthermore, unless otherwise clear from the context, a numerical value presented herein has an implied precision given by the least significant digit. Thus a value 1.1 implies a value from 1.05 to 1.15. The term “about” is used to indicate a broader range centered on the given value, and unless otherwise clear from the context implies a broader range around the least significant digit, such as “about 1.1” implies a range from 1.0 to 1.2. If the least significant digit is unclear,

then the term “about” implies a factor of two, e.g., “about X” implies a value in the range from $0.5\times$ to $2\times$, for example, about 100 implies a value in a range from 50 to 200. Moreover, all ranges disclosed herein are to be understood to encompass any and all sub-ranges subsumed therein. For example, a range of “less than 10” for a positive parameter can include any and all sub-ranges between (and including) the minimum value of zero and the maximum value of 10, that is, any and all sub-ranges having a minimum value of equal to or greater than zero and a maximum value of equal to or less than 10, e.g., 1 to 4.

(23) Some embodiments of the invention are described below in the context of tooth pulp vitality detection. However, the invention is not limited to this context. In other embodiments, the invention can be employed in the detection of blood vessels in open surgical and laparoscopic procedures or localization of blood vessels in the brain during electrode insertion.

(24) FIG. 1A through FIG. 1C are block diagrams that illustrate an example method to produce an interference signal at a detector, which indicates motion in a target volume. FIG. 1A is a block diagram that illustrates a Doppler shift in frequency and wavelength. A source **150**, such as an emitter or scatterer of a wave, such as an electromagnetic wave, including light, is moving in direction **152** relative to an observer. The waves **154** propagating in the direction of movement have an increased frequency and reduced wavelength compared to that emitted or scattered from a stationary source. Similarly, the waves **156** propagating in the direction opposite to the movement have a decreased frequency and increased wavelength compared to that emitted or scattered from a stationary source. This shift in frequency and wavelength is called a Doppler shift and is proportional to the speed v of the source **150** relative to the observer. For light waves traveling at the speed of light in the medium, given by c/N where c is the speed of light in a vacuum and N is the index of refraction of the medium, the Doppler shift $\Delta\lambda$ in original wavelength $\lambda_{sub.0}$ is given by Equation 1a and frequency shift by Equation 1b for an approaching source **150**.

$$\Delta\lambda = -(v/cN)\lambda_{sub.0} \quad (1a)$$

$$\Delta f = (v/cN)f_{sub.0} \quad (1b)$$

The optical wavelengths are very small, occurring in a range of hundreds of nanometers (nm, $1\text{ nm}=10^{-9}$ meters) and the corresponding frequencies are high on the order of hundreds of terahertz (THz, $1\text{ THz}=10^{12}$ Hertz).

(25) For many applications, such as medical applications with blood flow on the order of a meter per second, the speed v is a tiny fraction of the speed of light in the medium and so the Doppler shift is exceedingly small, on the order of 10^{-8} of the wavelength λ or frequency f . Such a change is difficult to measure directly. However, if the Doppler shifted wave interacts with a non-Doppler shifted wave, the interference pattern produces a signal related to the difference between the two waves that is much easier to measure. For example, a speed on the order of 1 meter per second (m/s) for a wavelength of about 1000 nm (f of about 300 THz) has a frequency shift of about 1 MegaHertz (MHz, $1\text{ MHz}=10^6$ Hertz, 1 Hertz, Hz, =1 cycle per second). FIG. 1B is a graph **160** that illustrates an example interference signal between two different waves. The horizontal axis **162** is time in arbitrary units, and the vertical axis **164** is amplitude in arbitrary units. The interference signal is given by trace **166**, which is enclosed by an envelope **168** of much lower frequency. Even for interfering optical waves with a frequency shift of one millionth of the optical frequencies, the interference signal envelope **168** has frequency on the order of MegaHertz. For flows of about a millimeters per second, the frequency is on the order of tens of kilohertz (kHz, $1\text{ KHz}=10^3$ Hertz).

(26) FIG. 1C is a block diagram that illustrates an example motion sensor system **100** based on optical measurements, as employed by many current motion sensor systems. A light source block **110** emits light of which one portion interacts with moving scatterers in block **114** and is Doppler shifted, and another portion is not Doppler shifted in block **112**, either because of interaction with stationary scatterers, or because this portion propagates in a reference path free from any scatterers, or some combination. The two paths are brought together to create an interference signal in block

120, which is detected and displayed in block **130**. How the different blocks are implemented varies widely in different devices. These types of existing motion sensor devices are described next: a laser Doppler flowmetry (LDF) device; a Doppler optical coherence tomography (DOCT) device; and a Coherence Gated Doppler Motion Sensing (CGD) system.

(27) FIG. 2A is a block diagram that illustrates an example laser Doppler flowmetry (LDF) apparatus **200** for detecting fluid flow within tissue. To illustrate the operation of device **200**, a sample **290** is depicted; however, sample **290** is not part of apparatus **200**. As used here and throughout, a sample refers to any subject on which a device operates, whether a living organism, including humans, animals and plants, or an excised portion thereof, or some other object, such as a device with moving parts, including moving fluids.

(28) The LDF apparatus **200** includes a narrow bandwidth laser source **210** called a single wavelength laser source, light from which is coupled to optical fiber **212a** that terminates in a first probe tip **214a**. This light is called emitted light and is signified in FIG. 2A by the letters “EM.” In practice, the laser source **210** is not absolutely single wavelength but comprises light in a very narrow wavelength band.

(29) The light from laser **210** exits the tip **214a** and interacts with a target volume **291** of the subject **290**. The target volume depends on the attenuation of the light at the single wavelength of laser **210** in the subject **290**. Light propagating through or scattered from the target volume, or both, enters tip **214b** of optical fiber **212b** and propagates along fiber **212b** from the tip to a detection circuit **230**. This light propagating through fiber **212b** is called returned light and is signified in FIG. 2A by the letters “RE.” The detection circuit outputs an electronic signal, such as current or voltage, proportional to the electric or magnetic field that impinges on a light sensitive portion of the detection circuit **230**. In the illustrated embodiment, the signal output by the detection circuit is an analog signal. That signal is passed to the analog to digital converter (ADC) **232** which outputs a digital signal that can be processed by a digital processor, as described below with reference to computer system **1000** in FIG. 10 or chip set **1100** in FIG. 11. Digital signal processing and display components **234** are configured to process the digital signal and presents the results, such as a graph or listing that indicates a number or speed of scatterers in the target volume **291** for one or more speed intervals at one or more times.

(30) The interference at the detection circuit **230**, if any, is caused by a difference between the returns from scatterers in target volume **291** that are not moving and the Doppler shifted returns from moving scatterers in target volume **291**. Because the same target volume **291** provides both the Doppler shifted and non-Doppler shifted optical waves (also referred to as photons, the quantum unit of electromagnetic waves), the interference pattern at the detection circuit **230** is said to be homodyne interference. While suitable for many purposes, the target volume is large compared to the features of interest (e.g., blood vessels or microchannels) in many applications. Thus the device **200** indicates whether there is motion in the volume **291** but does not indicate where in the volume the motion occurs. For example, one can determine that blood is flowing through tissue in the target volume but the blood vessels cannot be localized. Also, if there are too many background moving scatterers outside a region of interest or too few non-moving scatterers in the volume, the signal to noise ratio suffers and may render the measurements unreliable.

(31) The target volume **291** is difficult to control because of the long coherence length L_c of the narrowband laser **210**. Temporal coherence is a measure of the average correlation between the value of a wave and itself delayed by a time difference τ , at any pair of times and monochromaticity of the light source. In other words, it characterizes how well a wave can interfere with itself at a different time. The delay over which the phase or amplitude wanders by a significant amount (and hence the correlation decreases by significant amount) is defined as the coherence time τ_c . At $\tau=0$, the degree of coherence is perfect; whereas, it drops significantly by delay τ_c . The coherence length L_c is defined as the distance the wave travels in time τ_c . A wave with a longer coherence length is closer to a perfect sinusoidal wave. Wave interference is strong

when the paths taken by all of the interfering waves differ by less than the coherence length. Thus, with the long coherence length of the narrowband light source **210**, moving and non-moving scatterers throughout a large target volume **291** interfere significantly at detection circuit **230**.

(32) To better localize the vessels holding moving fluids in a sample, such as tissue of a living organism, imaging systems have been developed, which use wideband optical sources with short coherence lengths to achieve a very small and well controlled target volume. Such a small target volume is not likely to include a feature of interest, however; so, a scanning mechanism is included to scan a large number of target volumes, which can be presented as pixels or voxels in two dimensional (2D) or three dimensional (3D) images, respectively. One such imager is a DOCT device that requires a light source with a large optical bandwidth (typical bandwidth $\geq 6\%$ of the center wavelength), a lateral scanning and depth scanning mechanism, and a visual display.

(33) FIG. 2B is a block diagram that illustrates an example Doppler optical coherence tomography (DOCT) apparatus **240** for imaging fluid flow within tissue. This example is taken from Z. P. Chen, T. E. Milner, S. Srinivas, X. J. Wang, A. Malekafzali, M. J. C. vanGemert, and J. S. Nelson, "Noninvasive imaging of in vivo blood flow velocity using optical Doppler tomography," *Optics Letters*, v22, pp1119-1121, 1997. To illustrate the operation of device **240**, a sample **292** is depicted; however, sample **292** is not part of apparatus **240**.

(34) The DOCT apparatus **240** includes a wide bandwidth light source **250**, such as a superluminescent diode (SLD) to produce power at the scattering frequency of red blood cells combined with a Helium-Neon (He—Ne) laser for indicating the target volume by a position of a He—Ne laser spot. A SLD combines the high power and brightness of laser diodes with the low coherence of conventional light-emitting diodes. Its emission band is in a range from about 5 nm to about 100 nm wide. Light from this source **250** is coupled by lenses **252** and 2×1 splitter **254** and 2×2 splitter **256** to optical fiber in a sample arm **260**. A probe tip at distal end of sample arm **260** directs light into a minute target volume **293** in an in vivo or in vitro sample **292**. The probe tip is controlled by a scanning mechanism to scan multiple target volumes **293** in the sample **292**. For example, Z. P. Chen et al. state, "Two-dimensional images are formed by sequential lateral scans at a constant horizontal velocity of 800 $\mu\text{m/s}$ (micrometers, μm , also called microns, 1 μm =10^{sup.}-6 meters, per second), followed by incremental probe movements (10 μm) in the vertical (axial) direction." The scattered light is returned through the same sample arm to 2×2 splitter **256** which sends a portion to photodetector **270**.

(35) Because the target volume **293** is so small and light is scattered diffusely, it is not likely to return sufficient numbers of Doppler shifted and non-Doppler shifted photons from moving and non-moving scatterers, respectively, to generate a useful homodyne interference signal. In this case a reference path is provided through a reference arm optical fiber **280** to provide non-Doppler shifted reference photons with the appropriate mean delay interference at photodetector **270**.

Because the non-Doppler shifted photons are not from the same target volume **293** as the Doppler shifted photons, if any, this type of interference is called heterodyne interference. A portion of the light from source **250** is coupled into the reference arm by 2×2 splitter **256**. Because the signal from source **250** is broadband, its coherence length is very short and to achieve significant interference at photodetector **270** the differences in the path lengths between the returns from the sample arm **260** and the returns from the reference arm **280** must be less than the coherence distance L_c . To match the path distances, the reference arm includes a moveable mirror **286**. To avoid confounding homodyne interference with the desired heterodyne interference, the reference light is optionally changed in frequency by phase modulation component **282** so that the difference (and corresponding beat frequencies are much higher, about 10 kHz or greater) is greater than the maximum frequencies generated by blood flow. The reference arm light is directed to the photodetector by 2×2 splitter **256**. As stated by P. Chen et al., "Fluid-flow velocity at each pixel is determined by measurement of the Doppler frequency shift, which is defined as the difference between the carrier frequency established by the optical phase modulation and the centroid of the

measured power spectrum at each pixel.”

(36) The photodetector **270** outputs an electronic signal, such as current or voltage, proportional to the electric or magnetic field that impinges on the photodetector **270**. In the illustrated embodiment, the signal output by the photodetector **270** is an analog signal. That signal is passed to the analog to digital converter (ADC) **272** which outputs a digital signal that can be processed by a digital processor. Digital signal processing and image construction components **274** are configured to processes the digital signal and presents the results, such as an image of the locations and amounts of fluid motion, or statistics of same.

(37) FIG. 2C is a block diagram that illustrates an example laser Doppler flowmetry (LDF) system **294** for detecting fluid flow within tooth pulp. The system **294** is similar to the LDF system **200** of FIG. 2A except the tips **214a**, **214b** are positioned at a top surface of the tooth and are spaced close together (e.g. within 0.5 mm). Based on this close spacing of the tips **214a**, **214b**, the difference in the optical path length from the tip **214a** to the pulp region **297** and back to the tip **214b** and from the tip **214a** to the gum tissue **295** and back to the tip **214b** is relatively small and less than the long coherence length of the laser. Consequently, signals from the pulp region **297** and gum tissue **295** interfere at the detector and thus the system **294** cannot distinguish fluid motion within the pulp region **297** from fluid motion within the gum tissue **295**. Since the top surface of the pulp region **297** of most adult teeth, except for the small incisors, is located at least 3-4 mm below from the upper tooth surface, only a small fraction of the incident light is Doppler shifted by moving red cells within the blood vessels of the pulp region **297**. The top surface of a receding tooth pulp region **297** can be located at or below the gum line, even farther from the top surface of the tooth than the gum tissue **295**. The signal to noise ratio of signals from the pulp region **297** is not sufficient to detect fluid motion within the pulp region **297**, as a result of motion artifacts received from the gum tissue **295**.

(38) The inventors recognized that the sensitivity of fiber optic probe in the system **294** to detecting pulp flow is low and depends on placement of the probe tips **214a**, **214b** relative to the pulp region **297** and gum tissue **295**. Specifically, the inventors recognized that the probe tips **214a**, **214b** need to be positioned such that the difference in the optical path from the tip **214a** to the pulp region **297** and back to the tip **214b** and from the tip **214a** to the gum tissue **295** and back to the tip **214b** is greater than the laser coherence length so that scattered signals from the gum tissue **295** do not interfere at the detector and thus fluid motion within the pulp region **297** is measured while motion artifacts from the gum tissue **295** are suppressed.

1. Overview

(39) The current embodiment includes methods and devices that increase the sensitivity of laser Doppler flowmetry (LDF) to detect pulp blood flow, while rejecting gum blood flow more effectively. In the current embodiment, a novel transmission-mode configuration with an optical reference channel that increases both the interferometric gain and the effective size of the detection aperture achieves a large sensitivity gain. Rejection of gum blood flow is achieved by using a specific optical probe geometry combined with a light source that has a tailored spectral emission profile. An embodiment of an optical tooth vitality testing system includes a user-friendly, low-cost probe with a detachable portion that can be steam sterilized between uses.

(40) Various embodiments are based on a transmission-mode optical configuration that, when combined with other features, mitigates the limitations of low sensitivity and susceptibility to false readings from blood flow in the gums. For example, one embodiment includes a motion sensor that is suitable as a real-time guidance device for medical and other applications.

(41) FIG. 3A is a block diagram that illustrates example components of a coherence gated Doppler (CGD) system **300**, according to various embodiments. To illustrate the operation of system **300**, a sample **390** is depicted; however, sample **390** is not part of system **300**. The system **300** includes a selected bandwidth optical source **302** that determines both a center wavelength in order to appropriately penetrate a sample **390** and detect scatterers that move with the motion to be

detected, and also the coherence length L_c appropriate for a target volume **392** appropriate for the application. In some embodiments, the optical source **302** has an adjustable bandwidth. In other embodiments, the optical source **302** has a fixed or non-adjustable bandwidth. For example, for a hand-held probe, an appropriate target volume **392** is on the order of a cubic millimeter, such as volumes with spot diameters in a range from about 0.1 millimeters (100 microns) to about 5 millimeters. In some embodiments, the sample **390** is a tooth and the target volume **392** is a pulp region within the tooth. In one embodiment, light outputted from the optical source **302** is referred to as output light.

(42) The overall coherence gated sample volume **392** is affected by the dimension and focusing the optical source **302** and detector apertures and the scattering coefficient of the sample. A vertical extent of the coherence gated sample volume **392** is determined, at least in part, by the coherence length L_c . In an embodiment, where the sample **390** is the tooth and the target volume **392** is tooth pulp, to obtain a sufficiently high sensitivity to blood flow in the tooth pulp, while minimizing sensitivity to long path lengths through gum tissue, a wavelength bandwidth of about 0.5 nm or in a range from about 0.25 nm to about 1 nm is advantageous. For good transmission through the tooth, a center wavelength in a range of about 1250 nm to about 1350 nm or about 1550 nm to about 1750 nm is desirable to minimize scattering and absorption losses; however, other wavelengths in the visible or near infrared band can be employed at the expense of signal strength. In another embodiment, inexpensive semiconductor light sources are available at about 1310 nm and thus this wavelength is a good choice with a 0.5 nm bandwidth that corresponds to about 0.04% of the center wavelength. For a shorter center wavelength, the percentage is higher. For many applications, it is advantageous for the bandwidth to be intermediate between that of LDF and DOCT, e.g., for bandwidths in a range from about 0.01% to about 5% of a center wavelength. In some applications, it is preferable for the bandwidth to be in a range from about 0.01% to about 0.10%. In embodiments directed to the detection of blood cells, it is desirable for the bandwidth to be in a range from about 0.02% to about 0.08%. Because the target volume **392** is controlled, at least in part, by the coherence length of the source **310**, the volume **392** is called a coherence gated target volume **392** and the system **300** is called a coherence gated Doppler (CGD) device.

(43) The present system is designed to maximize light penetration and to control the size and location of the detection zone. Different types of light sources having various wavelengths may be used. Suitable light sources with selected center wavelength and bandwidth are known in the art. In one embodiment for application to teeth, a Superluminescent Light Emitting Diodes light source (SLED) with a center wavelength of about 1310 nm is used. This differs significantly from a conventional laser Doppler's wavelength of 780 nm because the teeth will not attenuate the longer wavelength infrared light as much. The SLED is also a preferred embodiment because it has a special type of laser diode with wider spectral bandwidth. As stated above, the bandwidth controls the coherence length which in turn controls the detection volume. Besides SLED, it is also possible to use a white light laser with suitable bandpass filter or semiconductor laser with appropriate bandwidth.

(44) The device also includes an optical detector **310** that outputs an electronic signal, such as a digital or analog current or voltage, which indicates the optical interference pattern at the detector. Example detectors include a photodiode, an Indium gallium arsenide (InGaAs) photodetector, a balanced detector, a charge-coupled device (CCD) and a Complementary metal-oxide-semiconductor (CMOS) photodetector. In some embodiments, a dual balanced detector is employed as detector **310** to improve signal to noise ratio by reducing common mode noise.

(45) The output light emitted from source **302** is directed into a probe **350** including tips **333a**, **333b** that engage opposite sides of the sample **390**. A first tip **333a** of the probe **350** engages a first side of the sample **390** and a second tip **333b** of the probe **350** engages a second opposite side of the sample **390**. In an embodiment, a pair of fiber optic lines pass through an interior of the probe **350** and include a source line **352** that is a fiber optic line which terminates at a distal end in the tip

333a. In an embodiment, the fiber optic line serving as the source line **352** is single mode or predominantly single mode. In other embodiments, the fiber optic line serving as the source line **352** is multi mode. In various embodiments, the distal end of the source line **352** is a simple perpendicular cut of the optical fiber serving as the source line **352** or an angled cut of the fiber serving as the source line **352** to control the back reflection power from the end surface, or a shaped end of the fiber to control the back reflection power from the end surface, or a coated end surface of the fiber to control the back reflection power from the end surface, or a Gradient-index (GRIN) multimode fiber to serve as a lens, or a GRIN rod lens, or ball lens, or a micro-lensed fiber or is a stiff needle connected to the source line **352**, or includes a lens or collimator, or a coating or other optical coupler, or some combination. The distal end of the source line **352** emits output light from source **302** and directs it onto a spot in or on the sample **390**. In some embodiments, an angled surface within the tip **333a** of the probe **350** redirects the output light emitted from the source line **352** in the sample **390**. In some embodiments, the probe **350** is configured to be held in a hand of a user or clamped on the surface of the tooth.

(46) A source line connector **353** is an optical connector that is configured to couple light received directly or indirectly (e.g., through optical coupler **304**) from the source **302** into the source line **352**. As used herein the term connector refers to any components or configuration of components that serve to direct light from one region (point, area or volume) of space to another region of space, and includes, among others, a free space vacuum, a gas or liquid filled free space, a crystal, a lens, Gradient-index (GRIN) optical component, such as a GRIN lens, a fiber bundle, a fiber coupler (FC), a beam splitter, a circulator, an optical fiber, a mirror, or any combination.

(47) The pair of fiber optic lines in the probe **350** also includes a detector line **354** that is a fiber optic line which terminates at a distal end in the tip **333b**. In an embodiment, the fiber optic line serving as the detector line **354** is multi mode or predominantly multi mode. In another embodiment, the fiber optic line serving as the detector line **354** is single mode. In various embodiments, the distal end of the detector line **354** is a simple perpendicular cut of the optical fiber serving as the detector line **354** or an angled cut of the fiber serving as the detector line **354** or a coated end surface of the fiber or a Gradient-index (GRIN) multimode fiber to serve as a lens, or a GRIN rod lens or ball lens or a micro-lensed fiber or is a stiff needle connected to the detector line **354** or includes a lens or collimator or a coating or other optical coupler or some combination. In an embodiment, the distal end of the detector line **354** receives scattered light from the target volume **392** and directs the scattered light to the detector **310** through an optical coupler **308**.

(48) A detector line connector **355** is configured to couple scattered light from the detector line **354** directly or indirectly (e.g., through optical coupler **308**) into the detector **310**.

(49) The scattered light returned from sample **390** is captured by the detector line **354** at probe tip **333b**. As is the case for probe tip **333a**, in various embodiments, the probe tip **333b** can take various forms. In some embodiments, the probe tip **333b** is configured to be held in a hand of a user. It is advantageous if the probe tip **333b** and probe tip **333a** are configured to form opposing surfaces of the probe **350** (e.g. caliper tips).

(50) In some embodiments, heterodyne interference is utilized and a reference path is introduced to direct a portion of the output light from the source **302** onto the optical detector **310**. In such embodiments, optical coupler **304** is included to direct a portion of the output light from the source **302** to an adjustable delay line **306** in the reference path, to an optical coupler **308** and the optical detector **310**. The configuration of the fiber couplers of the systems depend on the purpose of the system. Variations in the distribution of the laser power between the emitted light path directed to the source line **352** and the reference path directed to the adjustable delay line **306** will be application specific. In one example embodiment, the optical coupler **304** distributes 10% of laser power to the adjustable delay line **306** and the other 90% to the source line **352**. In other example embodiments, the laser power distributed to the probe **350** can be between about 99% and about 1%. For medical use, the preferred power expelled by the probe tip **333a** should not exceed the

United States Food and Drug Administration (FDA) recommendation. The optical probe tip **333a** is capable of focusing the laser beam in a perpendicular direction at different sizes. In one example, the beam is focused to a spot size of 40 μm at a distance 1.5 mm away from the probe **350**. However, in other uses, the spot size and distance may be more or less. The size of the target volume in the axial direction of propagation is at least in part determined, as described, above, by the coherence length L_c .

(51) In some embodiments, the adjustable delay line **306** includes a path length correction component so that a zero delay plane is positioned at the target volume **392** within the sample **390**. The zero-delay plane is defined as the plane in the target volume **392** which has the same optical path length to detector **310** as the optical path length through the reference path (e.g. adjustable delay line **306**) to detector **310**. Since the signal from the target volume **392** (defined by the coherence length L_c) surrounding the zero-delay plane is higher than signal from other regions including non-target volume **393**, the position of the zero-delay plane can be set by the reference path length to highlight the target volume **392** within the sample **390**. In some embodiments, the reference path length correction is adjustable so that the depth of the target volume **392** within the sample **390** can be adjusted. In an embodiment, the path is preset for a predetermined depth within the sample **390**. This embodiment is advantageous because the user does not have to make the adjustment, which renders the device simpler to use and frees the user's hands for other actions. Furthermore, the mechanism to adjust the path length can be omitted, making the device more stable, less susceptible to vibration or orientation, smaller, lighter, and cheaper to build.

(52) Electronic output from detector **310** is passed to a processor **312** to measure motion (e.g. flow of blood cells) within the target volume **392** based on a Doppler shift in a signal received at the detector **310** from the target volume **392** relative to the signal received at the detector **310** from the reference path. In some embodiments, an analog-to-digital converter (ADC) is positioned between the detector **310** and processor **312** to convert an analog signal from the detector **310** into a digital signal that is received by the processor **312**. The processor **312** includes a motion measurement process module **313** to perform one or more steps of a method described below with reference to FIG. **4**. In various embodiments, the processor **312** comprises one or more general purpose computer systems, as depicted in FIG. **10** or one or more chip sets as depicted in FIG. **11** and instructions to cause the computer or chip set to perform one or more steps of a method described below with reference to FIG. **4**. In some embodiments, an ADC is not used and an analog output from the detectors is presented at an analog device, such as an acoustic speaker. In such embodiments, a trained operator can determine qualitatively the blood flow within the pulp, and the health of the tooth, based on the frequency of the sound heard from the speaker.

(53) In some embodiments, the system **300** includes a presentation device **345** that indicates to a user the motion (amount of scatterers, speed of scatterers, or direction of scatterers, or some combination) in a coherence gated target volume **392** of sample **390**. Any presentation device may be used. In some embodiments, the presentation device **345** includes a digital port with digital output based on the electronic signal from optical detector **310**, which digital output may be used by the processor **312** (e.g. computer system **1000**) to display or speak one or more values that characterize the motion in target volume **392**. In some embodiments, the presentation device **345** is an acoustic speaker that emits an acoustic signal that depends on the output of optical detector **310**.

(54) In some embodiments direction of flow is inferred by rotating the probe tip in the vicinity of the sample. As the probe tip encounters more of the approaching flow, the frequency should increase; and, conversely as the probe tip encounters more of a receding flow the frequency should decrease. Thus, a direction of flow can be determined, at least qualitatively, by the dependence of frequency detected on the angle of the probe tip.

(55) Some combinations of features described above reduce cost and improve sensitivity enough to allow detection of flow in blood vessels of various sizes using a handheld probe with no moving parts. In various embodiments, CGD is also less dependent on the relative angle between the probe

and the blood vessel because, unlike conventional Doppler flow imaging, interference over long optical paths generates fluctuating speckle patterns that can be detected audibly even when vessels are probed at a 90 degree angle.

(56) There are a number of applications that would benefit from the system described herein. The improvements outlined above, in various embodiments, enable the applications of CGD to detection of blood vessels during instrument insertion for stereotactic neurosurgery; detection and avoidance of large blood vessels during anesthesia injection, catheterization, cerebrospinal fluid collection procedures, intravascular interventions or non-vascular interventions, vessel avoidance in the brain during deep brain electrode placement, tooth pulp vitality detection (with rejection of adjacent flow in the gums), and verification of flow cessation in ligated vessels during open surgical procedures.

(57) These devices and systems are not limited to medical use. The sensors may be designed to be used with engines or other instruments, such as microfluidic devices and micro-electro-mechanical systems (MEMS), where it is important to monitor or detect flow.

(58) FIG. 3B is a block diagram that illustrates example components of an apparatus used in the system 300 of FIG. 3A, according to various embodiments. In an embodiment, the apparatus is a probe 351 that is used as the probe 350 in the system 300. In an embodiment, the probe 351 takes the form of dental extracting forceps. The probe 351 includes a handle 356, a pivot 358 and a caliper 360, where the handle 356 and caliper 360 are pivotally rotatable about the pivot 358. In one embodiment, the handle 356 and caliper 360 are integral.

(59) In one embodiment, the source line 352 passes through an interior of the handle 356, around or through the pivot 358, and through the caliper 360. In another embodiment, the source line 352 passes along an exterior surface of one side of the handle 356, an exterior surface of the pivot 358 and over an exterior surface of the caliper 360 to the tip 333. In an embodiment, the source line 352 is single mode or predominantly single mode. In one embodiment, the source line 352 is single mode at least in the handle 356. The source line connector 353 is positioned at the end of the source line 352 and is configured to receive output light from the optical source 302 (through optical coupler 304). In an example embodiment, output light from the optical source 302 is transmitted through a single mode fiber (SMF) to the connector 353 that is a single mode to single mode connector.

(60) In one embodiment, the Single Mode Fiber (SMF) is an optical fiber designed to carry light only directly down the fiber—the transverse mode. Modes are the possible solutions of the Helmholtz equation for waves, which is obtained by combining Maxwell's equations and the boundary conditions. These modes define the way the wave travels through space, i.e. how the wave is distributed in space. Waves can have the same mode but have different frequencies. This is the case in single-mode fibers, where we can have waves with different frequencies, but of the same mode, which means that they are distributed in space in the same way, and that gives us a single ray of light. In one embodiment, the SMF has a core diameter between 8 and 10.5 μm and a cladding diameter of 125 μm . There are a number of special types of single-mode optical fiber which have been chemically or physically altered to give special properties, such as dispersion-shifted fiber and nonzero dispersion-shifted fiber. Data rates are limited by polarization mode dispersion and chromatic dispersion.

(61) In another embodiment, the single mode to single mode connector is used to join single mode optical fibers where a connect/disconnect capability is required. In one embodiment, the single mode to single mode connector is a connector assembly. A connector assembly consists of an adapter and two connector plugs. Due to the sophisticated polishing and tuning procedures that may be incorporated into optical connector manufacturing, connectors are generally assembled onto the optical fiber in a supplier's manufacturing facility. However, the assembly and polishing operations involved can be performed in the field, for example to make cross-connect jumpers to size.

(62) The detector line **354** also passes through an interior of the handle **356**, around or through the pivot **358**, and through the caliper **360**. The detector line **354** is multi mode or predominantly multi mode fiber. In one embodiment, the detector line **354** is a 50- μm or 62.5 μm graded-index low-dispersion telecommunications fiber. The detector line connector **355** is an optical connector that is positioned at the end of the detector line **354** and is configured to transmit scattered light from the detector line **354** to the detector **310** (through optical coupler **308**). In an example embodiment, scattered light from the detector line **354** is transmitted to the optical coupler **308** over a multi mode fiber (MMF) that is connected to the connector **355**, such that the connector **355** is a multi mode to multi mode connector.

(63) In one embodiment, the Multi Mode Fiber (MMF) is a type of optical fiber mostly used for communication over short distances. Typical multi-mode links have data rates of 10 Mbit/s to 10 Gbit/s over link lengths of up to 600 meters (2000 feet). In an embodiment, the Multi-mode fiber has a fairly large core diameter that enables multiple light modes to be propagated and limits the maximum length of a transmission link because of modal dispersion. The main difference between multi-mode and single-mode optical fiber is that the former has much larger core diameter, typically 50-100 micrometers; much larger than the wavelength of the light carried in it. Because of the large core and also the possibility of large numerical aperture, multi-mode fiber has higher “light-gathering” capacity than single-mode fiber. In one embodiment, the Multi-mode fiber is described based on the core and cladding diameters. In one example embodiment, a 62.5/125 μm multi-mode fiber has a core size of 62.5 micrometres (μm) and a cladding diameter of 125 μm . The transition between the core and cladding can be sharp, which is called a step-index profile, or a gradual transition, which is called a graded-index profile. The two types have different dispersion characteristics and thus different effective propagation distance. Multi-mode fibers may be constructed with either graded or step-index profile.

(64) The tip **333** of the probe **351** has a thickness **362** that is sized such that output light from the source line **352** is emitted from the tip **333** within a threshold distance of an edge **335** of the tip **333**. In an embodiment, the threshold distance is about 0.5 mm or within a range from about 0 mm to about 0.7 mm of the edge **335** of the tip **333**. In an example embodiment, a tip separation measurement is provided between opposing sides of the handle **356** which indicates a separation distance of the opposing sides of the handle **356**. In an example embodiment, a separation distance indication on the tip separation measurement correlates with a thickness **362** of the tip **333** of the caliper **360**.

(65) FIG. 3C is a block diagram that illustrates example components of an apparatus used in the system **300** of FIG. 3A, according to various embodiments. In an embodiment, the apparatus is a probe **351'** that is used as the probe **350** in the system **300**. In an embodiment, the probe **351'** is similar to the probe **351** with the exception of the features discussed herein. The caliper **360'** of the probe **351'** is removably attached to the handle **356'**.

(66) In one embodiment, the probe **351'** is configured to measure pulp flow in a variety of tooth types, and for convenient and low-cost use in dental practice. In some embodiments, the probe **351'** is fabricated from metal, ceramic, or a high-temperature plastic (e.g., PEEKTM from Curbell Plastics, Orchard Park, NY). In an embodiment, the probe **351'** includes a tapered tip, body, handles, and connectors. The tapered tip has a shape that is similar to a typical tooth extractor, for example. In one embodiment, the tapered tip is configured to reduce measured motion of the tapered tip relative to the tooth. For example, the tapered tip may be configured to clamp on the tooth or be temporarily fixed to a tooth by an adhesive, or the like.

(67) In an embodiment, the source line **352'** includes a single mode fiber (SMF) **370a** in the handle **356'** and a multi mode fiber (MMF) **372a** in the removable caliper **360'**. A single mode to multi-mode fiber connector **374a** is positioned between the caliper **360'** and the handle **356'** and is used to couple light from the SMF **370a** in the handle **356'** to the MMF **372a** in the caliper **360'** when the caliper **360'** is attached to the handle **356'**. The connector **374a** advantageously features a single

mode to multi mode connector, in order to loosen alignment tolerance and therefore, maximize the amount of light that is coupled from the source line **352'** in the handle **356'** to the source line **352'** in the caliper **360'**. This connector **374a** advantageously permits the tip **333'** and caliper **360'** to be detachable from the handle **356'**. The removable tip **333'** and caliper **360'** facilitate sterilization or disposal of the tip **333'** between uses. In one embodiment, the tip **333'** is disposable after use and replaced with a new sterilized tip **333'** for subsequent use. In another embodiment, the removable tip **333'** is sterilized using a sterilized solution or autoclave for a predetermined time period, as appreciated by one of ordinary skill in the art. After the removable tip **333'** is sterilized it is replaced on the handle **356'** and reused. In the illustrated embodiment the pivot **358'** is included in the removable tip **333'**. In some embodiments, the pivot **358'** is included in the reusable handle **356'** and not in the removable tip **333'**, and two removable tips are used—one for the source line caliper and one for the detector line caliper.

(68) In an embodiment, the detector line **354'** includes a multi mode fiber (MMF) **370b** in the handle **356'** and a multi mode fiber (MMF) **372b** in the caliper **360'**. A multi mode to multi mode connector **374b** is positioned between the caliper **360'** and the handle **356'** and is used to couple light from the MMF **370b** in the handle **356'** to the MMF **372b** in the caliper **360'** when the caliper **360'** is attached to the handle **356'**.

(69) In one embodiment, the tip **333'** is configured to be removable and disposable. For example, in one embodiment the tip **333'** is low-cost such that replacement of each tip due to wear and sanitation or hygiene requirements is cost effective. In another embodiment, the tip **333'** is configured to be removable and disposable and includes the light source, such as an integrated photonic circuit or laser-on-a-chip, or the like.

(70) In an embodiment, the tip **333'** of the caliper **360'** is arranged so that output light from the source line **352'** is emitted from the tip **333'** at a predetermined angle relative to a longitudinal axis **376** of the tip **333'**. In one embodiment, the predetermined angle is about 90 degrees or in a range from about 80 degrees to about 100 degrees. In one embodiment, the longitudinal axis **376** is defined by the caliper **360'** at the tip **333'**. In one embodiment, the caliper **360'** includes an angled surface **377** at the tip **333'** that is oriented to reflect or redirect light from the source line **352'** at the predetermined angle relative to the longitudinal axis **376**. In an example embodiment, the angled surface **377** is formed by polishing the tips of the fibers at about 45 degrees or in a range from about 30 degrees to about 60 degrees relative to the longitudinal axis **376** so that the light is redirected by total internal reflection. Alternatively, a separate lens/reflector assembly can be fabricated from plastic, glass or other transparent ceramic. FIG. 3D is a block diagram that illustrates the example source line **352'** of the apparatus of FIG. 3C, according to various embodiments. In an embodiment, the tip of the optical fiber **357** forming the source line **352'** is melted and then polished at about 45 degrees relative to the longitudinal axis **376**. The angled surface **377** is a mirrored surface positioned on the end of the optical fiber **357** to redirect the light within the tip **333b'** into the tooth **551** at the predetermined angle relative to the longitudinal angle **376** of the tip **333b'**. In one embodiment, the tip **333a'** includes an optical fiber **357** that features an angled surface **377** at an end of the optical fiber **357** to redirect the scattered light into the detector line **354'** that is similar to the angled surface **377** of the source line **352'**.

(71) In one embodiment, the probe **351'** is fabricated of an inexpensive glass molding and polishing to withstand the harsh conditions of high-temperature sterilization. In an embodiment, the body of the probe **351'** includes hollow channels that embed the optical fibers **370**, **372**. The connectors **353**, **355** are integrally formed with the handle **356'** to allow a patient-contacting portion of the probe **351'** (e.g. tip **333'** or caliper **360'**) to be steam-sterilized between uses without damaging the optical fibers **370**, **372** and strain relief and protective plastic jacket **373**.

(72) In some embodiments, the single source line **352** or **352'** serves as both emission path to direct output light from the light source **302** onto the sample **390** through the tip **333** and return path to capture scattered light from the sample **390** through the tip **333** and direct it to the optical detector

310 through an optical circulator. In some of these embodiments, the probe tip **333**, **333'** has a single tip rather than a dual tip.

(73) FIG. 4 is a flowchart that illustrates an example of a method **400** for detecting tooth pulp vitality, according to various embodiments. Although steps are depicted in FIG. 4 as integral steps in a particular order for purposes of illustration, in other embodiments, one or more steps, or portions thereof, are performed in a different order, or overlapping in time, in series or in parallel, or are omitted, or one or more additional steps are added, or the method is changed in some combination of ways. In step **401**, opposite sides of the sample **390** are engaged with the tip **333a**, **333b** of the probe **350**. In an embodiment, the opposite sides of the sample **390** are engaged with the tips **333a**, **333b** of the probe **350** so that the optical path length difference between the tips **333a**, **333b** through the target volume **392** and between the tips **333a**, **333b** through the non-target volume **393** is greater than the coherence length of the optical source **302**. In another embodiment, the opposite sides of the sample **390** are engaged with the tips **333a**, **333b** so that the optical path length difference is greater than one half of the coherence length of the optical source **302**.

(74) In an embodiment, in step **401**, opposite sides of a tooth are engaged with the tip **333** of the probe **351** or the tip **333'** of the probe **351'**. FIG. 5A is a block diagram that illustrates an example of positioning of a tip **333** of the probe **351** on opposite sides of a tooth **551** in the system **300** of FIG. 3A, according to various embodiments. A first tip **333a** of the probe **351** engages a first surface of the tooth **551** and a second tip **333b** of the probe **351** engages a second surface of the tooth **551** that is opposite from the first surface. In an embodiment, the first tip **333a** and second tip **333b** engage the tooth **551** so that the source line **352** and detector line **354** are positioned within a minimum distance $L_{sub.2}$ of gum tissue **550** surrounding the tooth **551**. In an embodiment, the first tip **333a** and second tip **333b** engage the tooth **551** so that the source line **352** and detector line **354** are positioned within a minimum distance $L_{sub.3}$ of pulp region **562** within the tooth **551**. Although FIG. 5A depicts that the tip **333a** does not engage the gum tissue **550**, in some embodiments the tip **333a** engages the gum tissue **550** and presses the gum tissue **550**. In one embodiment, the minimum distance $L_{sub.2}$ is about 0.5 mm or in a range from about 0 mm to about 2 mm. Additionally, in another embodiment, contours of the tip **333a**, **333b** are shaped to conform well to contours of the tooth **551**.

(75) FIG. 5A depicts an optical path **572** of light between the tips **333a**, **333b** through the tooth **551** that scatters from the pulp region **562**. The optical path **572** includes an initial diffusion length **560** where light incident from the source line **352** propagates in a straight direction within the tooth **551** after which the light deviates and is scattered from the pulp region **562** after which the light propagates along the diffusion length **560** before exiting the tooth **551** and entering the detector line **354**.

(76) FIG. 5A also depicts an optical path **570** of light between the tips **333a**, **333b** through the tooth **551** that scatters from gum tissue **550** surrounding the tooth **551**. In one embodiment, for purposes of this description, gum tissue **550** is defined as including one or more of gingival tissue, gingival margin, and periodontal ligament. The optical path **570** includes the initial diffusion length **560** after which the light backscatters and exits the tooth **551** before being scattered by the gum tissue **550** back into the tooth **551** and entering the detector line **354**. In an embodiment, the optical path **570** through the gum tissue **550** is greater than the optical path **572** through the pulp region **562** since light backscatters to the gum tissue **550**, exits the tooth **551**, scatters from the gum tissue **550** and reenters the tooth **551** before entering the detector line **354**.

(77) FIG. 5B is a plot **580** that illustrates example traces indicating an optical path difference between pulp region and gum tissue for various spacing between the probe and gum tissue, according to various embodiments. The horizontal axis **590** is thickness **502** of the tooth **551** and represented in units of millimeters (mm). The vertical axis **592** is a difference between the optical gum path **570** and optical pulp path **572** and represented in units of millimeters (mm). For purposes of the traces in FIG. 5B, the diffusion length **560** is about 0.43 mm, the index of refraction of the

tooth **551** is about 1.55 and the minimum distance $L_{sub.2}$ of the gum tissue is about equal to the minimum distance $L_{sub.3}$ of the pulp region. Trace **582** shows the difference between the optical gum path **570** and optical pulp path **572** where the minimum distance $L_{sub.2}$ (separation between the tip **333a**, **333b** and gum tissue **550**) is about 0.5 mm Trace **584** shows the difference between the optical gum path **570** and optical pulp path **572** where the minimum distance $L_{sub.2}$ (separation between the tip **333a**, **333b** and gum tissue **550**) is about 1 mm Trace **586** shows the difference between the optical gum path **570** and optical pulp path **572** where the minimum distance $L_{sub.2}$ (separation between the tip **333a**, **333b** and gum tissue **550**) is about 2 mm Trace **586** shows the difference between the optical gum path **570** and optical pulp path **572** where the minimum distance $L_{sub.2}$ (separation between the tip **333a**, **333b** and gum tissue **550**) is about 3 mm. The traces **582**, **584**, **586**, **588** demonstrate that the difference between the optical gum path **570** and optical pulp path **572** across multiple tooth thicknesses **502** and multiple spacings ($L_{sub.2}$) between the tips **333a**, **333b** and gum tissue **550** is greater than about 1.5 mm.

(78) In an embodiment, in step **401** the tips **333a**, **333b** engage the surface of the tooth **551** so that the path difference between the optical gum path **570** and optical pulp path **572** is greater than the coherence length of the optical source **302** so that scattered signals from the gum tissue **550** do not interfere at the detector **310** and thus motion artifacts from the gum tissue **550** are suppressed. In an embodiment, in step **401** the tips **333a**, **333b** engage the surface of the tooth **551** so that the path difference between the optical gum path **570** and optical pulp path **572** is greater than one half of the coherence length of the optical source **302**. In an embodiment, in step **401** the tips **333a**, **333b** engage the surface of the tooth **551** at a separation ($L_{sub.2}$) of about 0.5 mm above the gum tissue **550** such that the minimum path difference between the optical gum path **570** and optical pulp path **572** (e.g. at least 1.5 mm) is greater than half of the coherence length of the optical source **302** (e.g. about 3 mm).

(79) In an embodiment, in step **402**, output light is emitted from the optical source **302** with an adjustable bandwidth. In other embodiments, output light is emitted from the optical source **302** with a fixed bandwidth. A first portion of the output light is coupled into the source line **352** through the source line connector **353** and a second portion of the output light is coupled into the reference path including the adjustable delay line **306**. In one embodiment, in step **402** the optical coupler **304** directs the first portion of output light (e.g. 90%) to the source line connector **353** and the second portion of output light (e.g. 10%) to the adjustable delay line **306** in the reference path. (80) In an embodiment, in step **404**, the second portion of output light is received at the detector **310** from the reference path (e.g. through the optical coupler **308**) and scattered light from the detector line **354** is received at the detector **310** from the detector line connector **355** (e.g. through the optical coupler **308**).

(81) In an embodiment, in step **406**, the adjustable delay line **306** is varied so that a zero delay plane is positioned in the target volume **392** (e.g. pulp region **562** of the tooth **551**). In an embodiment, the zero delay plane is defined such that the optical path of output light through the reference path to the detector **310** is equal to an optical path from the zero delay plane through the detector line **354** to the detector **310**. In one embodiment, in step **406** the processor **312** receives an interference signal from the detector **310** and analyzes the interference signal. Based on this analysis, the processor **312** transmits a signal to the adjustable delay line **306** to vary the zero delay plane to be within the target volume **392** or within a center of the target volume **392**. In an example embodiment, the magnitude of the interference signal in different frequency bands can be used to determine the mean optical path delay through the tooth **551**, as well as the delay at which the Doppler power from cardiac pulsations is maximized.

(82) In an embodiment, in step **408**, the bandwidth of the output light is adjusted to vary the coherence length of the output light from the optical source **302** so that scattered light from the target volume **392** (e.g. pulp region **562**) generates an interference signal at the detector **310** and scattered light from the non-target volume **393** (e.g. gum tissue **550**) does not generate an

interference signal or generates an interference signal less than a threshold intensity at the detector **310**. In one embodiment, in step **408** the processor **312** receives an interference signal from the detector **310** and analyzes the interference signal. Based on this analysis, the processor **312** transmits a signal to the optical source **302** to vary the bandwidth of the output light so that interference signals from the non-target volume **393** are suppressed. In one embodiment, coherence is maintained over a range of optical paths defined by:

$$(83) \quad R = \langle l \rangle \pm \frac{l_c}{2} \quad (2)$$

where R is the range of optical paths that generate interference signals at the detector **310**; $\langle l \rangle$ is the mean optical path from the zero delay plane to the detector **310** and $l_{\text{sub.c}}$ is the coherence length of the optical source **310**. In an embodiment, in step **408** the coherence length $l_{\text{sub.c}}$ is adjusted so that the ratio $l_{\text{sub.c}}/2$ is less than the path difference (e.g. about 1.5 mm) between the optical gum path **570** and the optical pulp path **572** so that scattered light from the gum tissue **550** is suppressed at the detector **310** and does not generate an interference signal. In an example embodiment, in step **408** the coherence length $l_{\text{sub.c}}$ is adjusted to be less than about 3 mm or in a range from less than about 2 mm to less than about 4 mm. In other embodiments, the bandwidth of the output light is fixed and thus the coherence length is preset so that the ratio $l_{\text{sub.c}}/2$ is less than the path difference between the optical gum path **570** and the optical pulp path **572** so that scattered light from the gum tissue **550** is suppressed at the detector **310**.

(84) In some embodiments, although other light sources, such as a superluminescent diode, with a very short coherence length can exclude undesired optical paths through the gum tissue **550**, these light sources generally generate insufficiently strong interference signals. In an example embodiment, if the coherence length of a light source is too short, the volume of tissue probed by photons that penetrate the pulp region **562** may be too small to generate detectable Doppler signals. In addition, to attain adequate sensitivity, the aperture of the detector fiber must be wide enough to collect diffusely forward-scattered light.

(85) In an embodiment, in step **410**, the processor **312** determines motion within the target volume **392** (e.g. pulp region **562**) based on the interference signal received from the detector **310**. In one embodiment, in step **410** the processor **312** determines a flow of blood cells within the target volume **392** based on a Doppler shift between the scattered light from the pulp region **562** and the output light through the reference path.

(86) In yet another embodiment, the method **400** includes sterilizing the probe **351**, **351'** between uses or between use on different subjects. In one embodiment, for the probe **351**, the tip **333** is sterilized by positioning the tip **333** in a sterilized solution for a fixed amount of time. In yet another embodiment, for the probe **351'** with the replaceable caliper **360'**, the replaceable tip **333'** is detached from the handle **356'** and either disposed or sterilized using the same sterilization process as the tip **333**.

2. Example Embodiments

(87) FIG. **6** is a block diagram that illustrates example components of a coherence gated Doppler (CGD) system **600** for detecting tooth pulp vitality, according to various embodiments. In an embodiment, the system **600** is similar to the system **300** with the exception of the features discussed herein. In an embodiment, a single mode (SM) coupler **604** is provided that receives output light from the optical source **302** and directs a first portion (e.g. 90-99%) into a single mode fiber (SMF) which couples the first portion of output light into the source line **352**. The single mode coupler **604** also directs a second portion (e.g. 1-10%) into a SMF of the reference path and to the adjustable delay line **306**. In an example embodiment, the single mode coupler **604** is a 2×1 single mode coupler.

(88) In an embodiment, the source line **352** features a fiber tip reflector **653** at the tip **333a** to direct the output light from the source line **352** into the tooth **551** at the predetermined angle (e.g. about 90 degrees) relative to the longitudinal axis **376**. In another embodiment, the detector line **354**

features a fiber tip reflector **655** at the tip **333b** to receive the scattered light from the tooth **551** into the detector line **354**.

(89) In an embodiment, a multi mode coupler **608** is provided that receives output light from the adjustable delay line **306** through a multi mode fiber (MMF) and receives scattered light from the detector line **354** through a MMF. The multi mode coupler **608** directs the received output light and scattered light to a pair of balanced photodiodes **610** where the light from the reference path and scattered light from the detector line **354** generate an interference signal. One advantage of using the pair of balanced photodiodes **610** is a reduction of noise from the optical source **302** in generation of the interference signal. In an example embodiment, the multi mode (MM) coupler **608** is a 2×2 multi mode coupler.

(90) FIG. 7 is a block diagram that illustrates example components of a coherence gated Doppler (CGD) system **700** for detecting tooth pulp vitality, according to various embodiments. The system **700** is similar to the systems **300**, **600** with the exception of the features discussed herein.

(91) In step **402**, the optical source **302** emits output light to a Mach-Zehnder Interferometer **405**. In an embodiment, the output light enters the Mach-Zehnder Interferometer **405** and passes through an optical isolator **707** and to the Single Mode (SM) coupler **604**. The first portion of output light (e.g. 90-99%) is directed from the SM coupler **604** and into a SM fiber **720** through a connector **719a**. In an example embodiment, the connector **719a** is a single mode to single mode connector. The first portion of output light is transmitted through the SM fiber **720** and into the source line **352** through the connector **353**. In some embodiments, the SM fiber **720** should be long enough (e.g. 1-2 meters) to reach the connector **719a**.

(92) In an embodiment, the second portion of output light (e.g. 1-10%) is directed from the SM coupler **604** through a Single Mode (SM) fiber **721** and into the adjustable delay line **306**. Light from the adjustable delay line **306** is directed into a Multi Mode (MM) fiber **724** and subsequently into the MM coupler **608**. Additionally, scattered light from the detector line **354** is directed into a Multi Mode (MM) fiber **722** and into the MM coupler **608** through a connector **719b**. In an example embodiment, the connector **719b** is a Multi Mode to Multi Mode connector. In some embodiments, the MM fiber **722** should be long enough (e.g. 1-2 meters) to reach the connector **719b**. The MM coupler **608** directs the output light from the adjustable delay line **306** and scattered light from the detector line **354** to the pair of balanced photodetectors **710** using a pair of MM fibers. In an embodiment, the MM fibers used in the system **700** are graded-index fibers with a 62.5 micron or 50 micron core. However, other fiber types known to practitioners of the art can be employed. One advantage of the MM fibers **722** that couple the detector line **354'** to the MM coupler **608** and the MM fibers that direct the scattered light from the MM coupler **608** to the photodetectors **710** is that a large sample of scattered light is received from the pulp region **562** at the photodetectors **710** and thus the signal to noise ratio of scattered light from the pulp region **562** received at the photodetectors **710** exceeds a detection threshold.

(93) In an example embodiment, the photodetectors **710** include a photodiode, an indium gallium arsenide (InGaAs) photodetector, a balanced detector, a charge-coupled device (CCD) and a Complementary metal-oxide-semiconductor (CMOS) photodetector. In some embodiments, a dual-balanced detection circuit is employed as a detector to improve signal-to-noise ratio by reducing common-mode noise.

(94) In an embodiment, in step **404** light from the MM coupler **608** is received at the photodetectors **710** including output light from the reference path and scattered light from the detector line **354**. The photodetectors **710** output an electronic signal, such as current or voltage, proportional to the electric or magnetic field that impinges the photodetectors **710**. The signal output by the photodetectors **710** is passed to various components including a Direct Current (DC) amp **732**, an Alternating Current (AC) amp **734** and/or an Analog to Digital Converter (ADC) **736** before a digital signal is transmitted to the processor **312**. In an embodiment, the processor **312** analyzes the digital signal to determine whether to adjust the zero delay plane of the adjustable

delay line **306**. In step **406**, if the processor **312** determines that the zero delay plane needs to be adjusted, a delay line control **726** signal is transmitted to the adjustable delay line **306** so that the zero delay plane is positioned in the pulp region **562**. In one embodiment, in step **406** adjustable delay line **306** is adjusted automatically by the embedded processor **312** with the control signal **726** to maximize the interferometric signal strength generated by the pulp blood flow. In an example embodiment, either analog or digital signal processing, for example by the processor **312**, is employed to emphasize the pulsatile component of the pulp flow for indication, including audio and/or visual indications.

(95) In some of these embodiments, the optical interference signal at detectors **710** have high amplitude in frequencies between about 20 Hz and 20,000 Hz, and an analog output from the detectors **710** can directly drive an acoustic speaker **742** through a Digital to Analog converter **740** without the need for intervening significant processing or an intervening analog to digital converter (ADC). Acoustic output by the speakers **742** in this frequency range is audible to most humans. This enables the user to hear the difference in velocity as hand held probe tips **333a** and **333b** are moved across the tooth **551**. Vessels associated with moving fluid can be associated with visual features determined while the user's eyes are trained on the tooth **551**. This makes such embodiments suitable for real time guidance while keeping the system **700** light, simple and inexpensive, compared to both LDF and, especially, DOCT devices.

(96) In some of these embodiments, a wireless transmitter **738** is provided to transmit the digital signal from the ADC **736** to a remote location where a remote processor or display **744** is located to process the signal in a manner similar to the processor **312**. In some embodiments, the light source **302**, such as a laser, and other electro-optical components, including an optical delay line **306**, microcontroller **312**, and a battery, of the optical tooth vitality testing system **700** are housed in a compact unit with wireless connectivity to a smart tablet or smart phone, for example, for separate processing and display **744**. In other embodiments, the optical tooth vitality testing system **700** can be housed within a single portable instrument with wired connections among a pre-processor, processor, and display.

(97) In an embodiment, in step **408** the processor **312** adjusts the bandwidth of the optical source **302** to vary the coherence length of the output light from the optical source **302**. In one embodiment, the processor **312** transmits a signal through the DC amp **732** and Automatic Gain Control (AGC) **728** to a current source **730** to vary a current of the optical source **302** in order to adjust the bandwidth.

(98) In one embodiment, a feedback loop adjusts the intensity of the light source **302** to accommodate a wide range of teeth with different attenuation characteristics. In one embodiment, in step **408** the processor **312** adjusts the current of the optical source **302** in response to an intensity of the scattered light received by the photodetectors **310** so that the intensity of the scattered light is adjusted to be above a threshold intensity. This advantageously accommodates a range of tooth thickness **502** that can affect the intensity of the scattered light received by the photodetectors **310** and correspondingly adjusts the intensity of the optical source **302** to ensure that a minimum threshold intensity of scattered light is received at the photodetectors **310** for a wide range of tooth thickness **502**.

(99) Most conventional semiconductor lasers do not have the properties required to achieve adequate sensitivity and selectivity to pulp blood flow. In an embodiment, the optical source **302** has one or more of the following characteristics: 1) a peak emission wavelength between about 1250 nm and about 1325 nm or between about 1600 nm and about 1700 nm; 2) an output power greater than about 5 mW; 3) a linewidth between about 0.25 nm and about 1 nm and having a single lobe emission profile (rather than, for example, a comb-like emission profile); 4) compact form; and 5) low cost of manufacture and operation.

(100) In one embodiment, the optical source **302** includes a laser with custom-tailored coherence properties as light sources to satisfy. Mass-produced surface-mounted lasers, such as vertical-cavity

surface-emitting lasers (VCSELs), emit low optical power (less than 2 mW) in a single mode and have the emission wavelength in a shorter wavelength band (approximately between 800 nm and 980 nm). Since the magnitude of the Doppler signal is proportional to source power and scattering losses in the tooth decrease with wavelength, short-wavelength VCSELs generally fall short of the ideal light sources for tooth vitality testing. In some embodiments, VCSELs can be manufactured to operate at longer wavelengths with appropriate linewidths and emission power levels. Other types of laser and light sources are contemplated, including integrated photonic circuits and lasers-on-a-chip, for example.

(101) Although semiconductor Fabry-Perot (F-P) lasers are available with the desired power and peak wavelength characteristics, these lasers have a comb-like emission profile that produce multiple coherence time (or length) intervals. For example, a typical 1310 nm F-P laser has a narrow (e.g., less than 0.05 nm) main lobe and several equally narrow side lobes over a band of approximately between 1 nm and 2 nm, which generates a coherence function with multiple coherence intervals separated by approximately between 0.5 mm and 2 mm. In one embodiment, the coherence function with such a comb-like shape produces an undesired sensitivity to optical paths longer than the coherence function from the main lobe alone. These side lobes can be suppressed, but the main lobe remains too narrow to restrict the coherence length to less than a few millimeters.

(102) FIG. 8 is a block diagram that illustrates an example of a light source **802** used in the system **700** of FIG. 7, according to various embodiments. In an embodiment, the light source **802** is capable of generating output light with the desired emission characteristics discussed above. In an embodiment, the light source **802** includes a Fiber Bragg Grating (FBG) **804**, a Semiconductor Optical amplifier (SOA) **806** and an optical isolator **807**. In an embodiment, an amplified spontaneous emission (ASE) from one terminal of the fiber-coupled SOA **806** is coupled into the custom reflective FBG **804**. In an embodiment, the FBG **804** is configured to produce a smooth narrowband reflectance profile. Light is amplified by the SOA **806** in a single pass, is reflected by the FBG **804**, and is emitted at the output terminal. The optical isolator **807** minimizes the effects of spurious reflections at the output terminal. In one embodiment, the light source **802** can generate a single emission line of light with a full-width-at-half-maximum (FWHM) width approximately between about 0.25 nm and about 1 nm at an output power greater than about 10 mW. In practice, the light source **802** is not absolutely single wavelength but comprises light in a very narrow wavelength band, typically with a wavelength bandwidth of about 0.08% of a center wavelength in the band or less.

(103) FIG. 9A is a block diagram that illustrates an example of positioning a tip **333''** of a probe **351''** on opposite sides of a tooth **551** in the system of FIG. 7, according to various embodiments. In an embodiment, upon engagement of the tip **333''** on the opposing surfaces of the tooth **551**, an optical axis of the source line **352** in the tip **333''** is roughly aligned with an optical axis of the detector line **354** in the tip **333''**. In some embodiments, anatomic constraints may not permit precise optical axis alignment of the source line **352** and detector line **354**. In one example embodiment, it may be advantageous to direct the source line **352** and detector line **354** in the tip **333''** slightly towards the apex of the root rather than directly towards each other.

(104) In an embodiment, the tip **333''** is covered with a soft clear material with high transmission of light in the visible to the mid-infrared spectrum (e.g. silicone). However, in other embodiments, the tip **333''** excludes such soft clear material. In another embodiment, in step **401** the opposing sides of the tip **333''** engage the tooth **551** at a cemento-enamel junction (CEJ) **951** of the tooth **551** so that output light is emitted from the tip **333''** at or below the CEJ **951** and scattered light is received by the tip **333''** at or below the CEJ **951**. One advantage of engaging the tip **333''** at or below the CEJ of the tooth is that dentine tissue below the CEJ and has better light transmitting properties than the calcified crown tissue that is above the CEJ. Another advantage of engaging the tip **333''** at or below the CEJ is that an enamel layer, which is less transparent to the light is narrowed and/or absent below the CEJ **951**. Thus, attaching the tip **333''** at or below the CEJ **951** of the tooth **551**

enhances the signal to noise ratio of scattered light from the pulp region **562** received at the detector **710**. In yet another embodiment, in step **401** after the opposite sides of the tip **333** engage the CEJ of the tooth **551**, the optical fibers **372a**, **372b** at the tip **333** can be adjusted relative to the tip **333** in order to optimize position of the fibers **372a**, **372b** relative to the surface of the tooth **551**.

(105) In one embodiment, the probe **351** depicted in FIG. **9A** is smaller, lighter and simpler to build than the probe designs of FIGS. **3B** and **3C**. In one embodiment, the probe **351** is a disposable probe. In an embodiment, the fiber optic line **720** serving as the source line **352** is single mode. In one embodiment, the probe **351** is a single flexible unit with two caliper arms. In one example embodiment, the probe **351** has no handle or a much reduced handle as compared to the handles **356**, **356'** of the probes **351**, **351'**. In one example embodiment, the tip **333** is made from a spring like polymer. In another example embodiment, a spring is positioned adjacent the pivot of the probe **351**. In an embodiment, the tip **333** has a soft clear low slippage material. In an embodiment, the two caliper arms containing the source line **352** and detector line **354** are surrounded and protected by a spring like polymer that allow the caliper arms to be opened and closed around the tooth **551**. In some embodiments, the smaller probe **351** can be positioned and removed with a scissors like device that is used to open and close the calipers of the probe **351**. In an example embodiment, the scissors like device opens or closes the caliper arms of the probe **351** by positioning each arm of the scissors like device between the tooth **551** and each caliper arm to press the caliper arms outward (open) or guide the caliper arms inward (close).

(106) In one embodiment, in step **401** three contact points are established between the probe **351** and teeth of the subject to ensure that there is minimal movement during measurement. In one embodiment, a first contact point is established between a first side of the tip **333** and the tooth **551** for the delivery of the output light from the source line **352**. A second contact point is established between the second side of the tip **333** and the tooth **551** for the detection of scattered light at the detector line **354**. In an embodiment, the third contact is established to stabilize the probe tip **333** during engagement with the tooth **551**. In some embodiments, the third contact point is a large surface rather than the small surface of the former two contact points. In one embodiment, the third contact point is on top of the crown and orthogonal to the axis of the tooth **551**. In another embodiment, the third contact point is on neighboring teeth (e.g. crown of a neighboring tooth). In an embodiment, the first two contact points for engaging the tip **333** are positioned below the crown so that the forces from the three contact points oppose each other to effect mechanical stability. In an embodiment, the three contact points are covered with a stabilization contact **950** that is made from a thin non-rigid, clear, non-slip material to optimally hold the probe stationary on the tooth **551** and/or neighboring tooth.

(107) FIG. **9B** is a block diagram that illustrates an example of positioning the tip **333** of the probe **351** on opposite sides of a tooth **551** below the gum line **956** in the system **700** of FIG. **7**, according to various embodiments. In an embodiment, in step **401** the opposite sides of the tip **333** engages the gum line **956** and presses the gum tissue **957** adjacent the tooth **551** downward below the gum line **956**. In an example embodiment in step **401** the opposite sides of the tip **333** presses the gum tissue **957** downward in order to access the CEJ **951** so that the tip **333** is positioned adjacent (e.g. at or below) to the CEJ **951**. In an embodiment, in step **401** the opposite sides of the tip **333** engage the opposite sides of the tooth **551** below the gum line **956** after the gum tissue **957** adjacent the tooth **551** has been pressed down below the gum line **956**. One advantage of this embodiment is that the tip **333**, in addition to the source line **352** and detector line **354** are below an enamel layer **958** of the tooth **551** that is less transparent to output light from the optical source **302**. Another advantage of this embodiment is that dentin which is more transparent to output light is between the tip **333** and the pulp region **962**. In an example embodiment, in step **401** the tip **333** is positioned such that output light is emitted from the source line **352** and scattered light is received by the detector line **354** within about 0.5 mm of the gum tissue **957** after the gum tissue **957** has

been pressed down by the tip **333**. Thus, this arrangement advantageously enhances the signal to noise ratio of scattered light received from the pulp region **962**.

(108) In one embodiment, a spring **967** is positioned between opposite sides of the probe **351** in a vicinity of the pivot **358** to press opposing sides of the tip **333** together and thus maintain tip **333** engagement on opposing surfaces (e.g. buccal and lingual) of the tooth **551**.

(109) In an embodiment, the size and location of the pulp region **962** varies depending on the age of the subject. In one embodiment, for a younger subject the pulp region **962** extends well into the crown of the tooth **551**. In another embodiment, for an older subject the pulp region **962** retracts downwards from the crown of the tooth **551**.

(110) In one embodiment, undesired scattered light along the optical path **570** (FIG. 5A) from the gum tissue **550** is scattered from periodontal ligament tissue that surrounds the root of the tooth **551**. In yet another embodiment, the undesired scattered light is scattered from gingival tissue (e.g. gingival margin). The enamel layer **958** is highly calcified and strongly scatters the output light from the tip **333** whereas the dentine transmits the output light relatively well. In one embodiment, in step **401** the opposite sides of the tip **333** engage the tooth **551** at the cemento-enamel junction (where the enamel layer **958** has narrowed and merges with the cementum layer).

(111) Access to the cemento-enamel junction varies based on the age and/or gum disease of the subject. In some embodiments, the probe is designed with different sizes or features, depending on the age or gum disease stage of the subject. In one example embodiment, access to the cemento-enamel junction is limited in young subjects. However, the pulp region **962** typically extends above the cemento-enamel junction and up into the crown of the tooth **551** in young subjects and the tooth **551** is less calcified. Accordingly, in one embodiment, in step **401** the opposite sides of the tip **333** engage the tooth **551** above the cemento-enamel junction and in the crown region of the tooth **551**, for younger subjects. In another embodiment, in step **401** the opposite sides of the tip **333** engage the tooth **551** below the cemento-enamel junction for older subjects.

(112) FIG. 9C is a block diagram that illustrates an example of a tip **333** of the probe **351** of FIG. 7, according to various embodiments. In an embodiment, the tip **333** includes the MMF fiber **372** of the source line **352** that terminate at a Gradient Index-Optics (GRIN) lens **969** and features a soft clear low slippage contact material **965**. However, in some embodiments, the GRIN lens **969** is not used and is replaced with a different optical component. In another embodiment, the MMF fiber **372** terminates at a mirror that redirects output light from the source line **352** into the tooth **551** at the predetermined angle (e.g. about 90 degrees relative to the longitudinal axis **376**). In yet another embodiment, the MMF fiber **372** is positioned within a separation **980** of an edge **335** of the tip **333** such that output light is emitted from the tip **333** at the separation **980** from the edge **335** of the tip **333**. In an example embodiment, the separation **980** is equal to or less than a separation between the gum line **956** and the pulp region **962** (FIG. 9B). In an example embodiment, the optical fibers used in the source line **352** and/or detector line **354** are rated to transmit light with a radius of curvature in a range from about 5 mm to about 8 mm, which is sufficient for the curvature of the tip **333**.

(113) An advantage of the transmission mode of the probe tip **333** arrangement is that light is scattered stronger in a forward direction than compared to backscattering. Additionally, when a detector tip **333a** and emitter tip **333b** are in alignment, a spatial specificity can be improved for detecting signals from the target (e.g. pulp region **562**) placed between the detector and emitter. Placing the emitter and detector on opposite sides of a target volume is generally not feasible for most medical applications but the probe tip **333** is advantageously configured to stably clip onto the tooth **551** for use in dentistry. Another challenge with using dental probes when imaging teeth is ensuring sterility of the probes between patients or uses. Thus, the probe tip **333'** advantageously addresses this challenge by being detachable so that the tip **333'** can be either disposed or sterilized between patients.

3. Hardware Overview

(114) FIG. 10 is a block diagram that illustrates a computer system 1000 upon which an embodiment of the invention may be implemented. Computer system 1000 includes a communication mechanism such as a bus 1010 for passing information between other internal and external components of the computer system 1000. Information is represented as physical signals of a measurable phenomenon, typically electric voltages, but including, in other embodiments, such phenomena as magnetic, electromagnetic, pressure, chemical, molecular atomic and quantum interactions. For example, north and south magnetic fields, or a zero and non-zero electric voltage, represent two states (0, 1) of a binary digit (bit).). Other phenomena can represent digits of a higher base. A superposition of multiple simultaneous quantum states before measurement represents a quantum bit (qubit). A sequence of one or more digits constitutes digital data that is used to represent a number or code for a character. In some embodiments, information called analog data is represented by a near continuum of measurable values within a particular range. Computer system 1000, or a portion thereof, constitutes a means for performing one or more steps of one or more methods described herein.

(115) A sequence of binary digits constitutes digital data that is used to represent a number or code for a character. A bus 1010 includes many parallel conductors of information so that information is transferred quickly among devices coupled to the bus 1010. One or more processors 1002 for processing information are coupled with the bus 1010. A processor 1002 performs a set of operations on information. The set of operations include bringing information in from the bus 1010 and placing information on the bus 1010. The set of operations also typically include comparing two or more units of information, shifting positions of units of information, and combining two or more units of information, such as by addition or multiplication. A sequence of operations to be executed by the processor 1002 constitutes computer instructions.

(116) Computer system 1000 also includes a memory 1004 coupled to bus 1010. The memory 1004, such as a random access memory (RAM) or other dynamic storage device, stores information including computer instructions. Dynamic memory allows information stored therein to be changed by the computer system 1000. RAM allows a unit of information stored at a location called a memory address to be stored and retrieved independently of information at neighboring addresses. The memory 1004 is also used by the processor 1002 to store temporary values during execution of computer instructions. The computer system 1000 also includes a read only memory (ROM) 1006 or other static storage device coupled to the bus 1010 for storing static information, including instructions, that is not changed by the computer system 1000. Also coupled to bus 1010 is a non-volatile (persistent) storage device 1008, such as a magnetic disk or optical disk, for storing information, including instructions, that persists even when the computer system 1000 is turned off or otherwise loses power.

(117) Information, including instructions, is provided to the bus 1010 for use by the processor from an external input device 1012, such as a keyboard containing alphanumeric keys operated by a human user, or a sensor. A sensor detects conditions in its vicinity and transforms those detections into signals compatible with the signals used to represent information in computer system 1000. Other external devices coupled to bus 1010, used primarily for interacting with humans, include a display device 1014, such as a cathode ray tube (CRT) or a liquid crystal display (LCD), for presenting images, and a pointing device 1016, such as a mouse or a trackball or cursor direction keys, for controlling a position of a small cursor image presented on the display 1014 and issuing commands associated with graphical elements presented on the display 1014.

(118) In the illustrated embodiment, special purpose hardware, such as an application specific integrated circuit (IC) 1020, is coupled to bus 1010. The special purpose hardware is configured to perform operations not performed by processor 1002 quickly enough for special purposes. Examples of application specific ICs include graphics accelerator cards for generating images for display 1014, cryptographic boards for encrypting and decrypting messages sent over a network, speech recognition, and interfaces to special external devices, such as robotic arms and medical

scanning equipment that repeatedly perform some complex sequence of operations that are more efficiently implemented in hardware.

(119) Computer system **1000** also includes one or more instances of a communications interface **1070** coupled to bus **1010**. Communication interface **1070** provides a two-way communication coupling to a variety of external devices that operate with their own processors, such as printers, scanners and external disks. In general the coupling is with a network link **1078** that is connected to a local network **1080** to which a variety of external devices with their own processors are connected. For example, communication interface **1070** may be a parallel port or a serial port or a universal serial bus (USB) port on a personal computer. In some embodiments, communications interface **1070** is an integrated services digital network (ISDN) card or a digital subscriber line (DSL) card or a telephone modem that provides an information communication connection to a corresponding type of telephone line. In some embodiments, a communication interface **1070** is a cable modem that converts signals on bus **1010** into signals for a communication connection over a coaxial cable or into optical signals for a communication connection over a fiber optic cable. As another example, communications interface **1070** may be a local area network (LAN) card to provide a data communication connection to a compatible LAN, such as Ethernet. Wireless links may also be implemented. Carrier waves, such as acoustic waves and electromagnetic waves, including radio, optical and infrared waves travel through space without wires or cables. Signals include man-made variations in amplitude, frequency, phase, polarization or other physical properties of carrier waves. For wireless links, the communications interface **1070** sends and receives electrical, acoustic or electromagnetic signals, including infrared and optical signals, that carry information streams, such as digital data.

(120) The term computer-readable medium is used herein to refer to any medium that participates in providing information to processor **1002**, including instructions for execution. Such a medium may take many forms, including, but not limited to, non-volatile media, volatile media and transmission media. Non-volatile media include, for example, optical or magnetic disks, such as storage device **1008**. Volatile media include, for example, dynamic memory **1004**. Transmission media include, for example, coaxial cables, copper wire, fiber optic cables, and waves that travel through space without wires or cables, such as acoustic waves and electromagnetic waves, including radio, optical and infrared waves. The term computer-readable storage medium is used herein to refer to any medium that participates in providing information to processor **1002**, except for transmission media.

(121) Common forms of computer-readable media include, for example, a floppy disk, a flexible disk, a hard disk, a magnetic tape, or any other magnetic medium, a compact disk ROM (CD-ROM), a digital video disk (DVD) or any other optical medium, punch cards, paper tape, or any other physical medium with patterns of holes, a RAM, a programmable ROM (PROM), an erasable PROM (EPROM), a FLASH-EPROM, or any other memory chip or cartridge, a carrier wave, or any other medium from which a computer can read. The term non-transitory computer-readable storage medium is used herein to refer to any medium that participates in providing information to processor **1002**, except for carrier waves and other signals.

(122) Logic encoded in one or more tangible media includes one or both of processor instructions on a computer-readable storage media and special purpose hardware, such as ASIC *1020.

(123) Network link **1078** typically provides information communication through one or more networks to other devices that use or process the information. For example, network link **1078** may provide a connection through local network **1080** to a host computer **1082** or to equipment **1084** operated by an Internet Service Provider (ISP). ISP equipment **1084** in turn provides data communication services through the public, world-wide packet-switching communication network of networks now commonly referred to as the Internet **1090**. A computer called a server **1092** connected to the Internet provides a service in response to information received over the Internet. For example, server **1092** provides information representing video data for presentation at display

1014.

(124) The invention is related to the use of computer system **1000** for implementing the techniques described herein. According to one embodiment of the invention, those techniques are performed by computer system **1000** in response to processor **1002** executing one or more sequences of one or more instructions contained in memory **1004**. Such instructions, also called software and program code, may be read into memory **1004** from another computer-readable medium such as storage device **1008**. Execution of the sequences of instructions contained in memory **1004** causes processor **1002** to perform the method steps described herein. In alternative embodiments, hardware, such as application specific integrated circuit **1020**, may be used in place of or in combination with software to implement the invention. Thus, embodiments of the invention are not limited to any specific combination of hardware and software.

(125) The signals transmitted over network link **1078** and other networks through communications interface **1070**, carry information to and from computer system **1000**. Computer system **1000** can send and receive information, including program code, through the networks **1080**, **1090** among others, through network link **1078** and communications interface **1070**. In an example using the Internet **1090**, a server **1092** transmits program code for a particular application, requested by a message sent from computer **1000**, through Internet **1090**, ISP equipment **1084**, local network **1080** and communications interface **1070**. The received code may be executed by processor **1002** as it is received, or may be stored in storage device **1008** or other non-volatile storage for later execution, or both. In this manner, computer system **1000** may obtain application program code in the form of a signal on a carrier wave.

(126) Various forms of computer readable media may be involved in carrying one or more sequence of instructions or data or both to processor **1002** for execution. For example, instructions and data may initially be carried on a magnetic disk of a remote computer such as host **1082**. The remote computer loads the instructions and data into its dynamic memory and sends the instructions and data over a telephone line using a modem. A modem local to the computer system **1000** receives the instructions and data on a telephone line and uses an infra-red transmitter to convert the instructions and data to a signal on an infra-red a carrier wave serving as the network link **1078**. An infrared detector serving as communications interface **1070** receives the instructions and data carried in the infrared signal and places information representing the instructions and data onto bus **1010**. Bus **1010** carries the information to memory **1004** from which processor **1002** retrieves and executes the instructions using some of the data sent with the instructions. The instructions and data received in memory **1004** may optionally be stored on storage device **1008**, either before or after execution by the processor **1002**.

(127) FIG. **11** illustrates a chip set **1100** upon which an embodiment of the invention may be implemented. Chip set **1100** is programmed to perform one or more steps of a method described herein and includes, for instance, the processor and memory components described with respect to FIG. ***10** incorporated in one or more physical packages (e.g., chips). By way of example, a physical package includes an arrangement of one or more materials, components, and/or wires on a structural assembly (e.g., a baseboard) to provide one or more characteristics such as physical strength, conservation of size, and/or limitation of electrical interaction. It is contemplated that in certain embodiments the chip set can be implemented in a single chip. Chip set **1100**, or a portion thereof, constitutes a means for performing one or more steps of a method described herein.

(128) In one embodiment, the chip set **1100** includes a communication mechanism such as a bus **1101** for passing information among the components of the chip set **1100**. A processor **1103** has connectivity to the bus **1101** to execute instructions and process information stored in, for example, a memory **1105**. The processor **1103** may include one or more processing cores with each core configured to perform independently. A multi-core processor enables multiprocessing within a single physical package. Examples of a multi-core processor include two, four, eight, or greater numbers of processing cores. Alternatively or in addition, the processor **1103** may include one or

more microprocessors configured in tandem via the bus **1101** to enable independent execution of instructions, pipelining, and multithreading. The processor **1103** may also be accompanied with one or more specialized components to perform certain processing functions and tasks such as one or more digital signal processors (DSP) **1107**, or one or more application-specific integrated circuits (ASIC) **1109**. A DSP **1107** typically is configured to process real-world signals (e.g., sound) in real time independently of the processor **1103**. Similarly, an ASIC **1109** can be configured to performed specialized functions not easily performed by a general purposed processor. Other specialized components to aid in performing the inventive functions described herein include one or more field programmable gate arrays (FPGA) (not shown), one or more controllers (not shown), or one or more other special-purpose computer chips.

(129) The processor **1103** and accompanying components have connectivity to the memory **1105** via the bus **1101**. The memory **1105** includes both dynamic memory (e.g., RAM, magnetic disk, writable optical disk, etc.) and static memory (e.g., ROM, CD-ROM, etc.) for storing executable instructions that when executed perform one or more steps of a method described herein. The memory **1105** also stores the data associated with or generated by the execution of one or more steps of the methods described herein.

(130) In the foregoing specification, the invention has been described with reference to specific embodiments thereof. It will, however, be evident that various modifications and changes may be made thereto without departing from the broader spirit and scope of the invention. The specification and drawings are, accordingly, to be regarded in an illustrative rather than a restrictive sense. Throughout this specification and the claims, unless the context requires otherwise, the word “comprise” and its variations, such as “comprises” and “comprising,” will be understood to imply the inclusion of a stated item, element or step or group of items, elements or steps but not the exclusion of any other item, element or step or group of items, elements or steps. Furthermore, the indefinite article “a” or “an” is meant to indicate one or more of the item, element or step modified by the article. As used herein, unless otherwise clear from the context, a value is “about” another value if it is within a factor of two (twice or half) of the other value. While example ranges are given, unless otherwise clear from the context, any contained ranges are also intended in various embodiments. Thus, a range from 0 to 10 includes the range 1 to 4 in some embodiments.

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Claims

1. An apparatus comprising: a handle; a pivot; a caliper; and a pair of fiber optic lines passing through an interior of the handle, the pivot and the caliper; wherein one of the fiber optic lines is a source line with a connector at an end of the source line and wherein the source line is single mode in the handle; wherein another of the fiber optic lines is a detector line with a connector at an end of the detector line and wherein the detector line is multi mode; wherein the caliper includes a tip configured to engage opposite sides of a sample; wherein the source line is configured to receive light from an optical source having a coherence length; and wherein the tip is configured such that a difference between a first optical path along the source line and the detector line through a target volume of the sample and a second optical path along the source line and the detector line through a non-target volume of the sample exceeds the coherence length of the light to optically suppress signals from the non-target volume at the detector line such that the signals from the non-target volume do not interfere at the detector line and such that motion within the target volume is measured while motion within the non-target volume is suppressed.
2. An apparatus as recited in claim 1, wherein the caliper is removable and wherein the source line is multi mode in the caliper.
3. An apparatus as recited in claim 2, wherein the source line includes a single mode to multi mode connector between the handle and the caliper and wherein the detector line includes a multi mode to multi mode connector between the handle and the caliper.
4. An apparatus as recited in claim 1, wherein a thickness of the tip of the caliper is sized such that the light from the source line is configured to be emitted from the tip within about 0.5 mm of an edge of the tip.
5. An apparatus as recited in claim 1, wherein the tip of the caliper is configured such that the light from the source line is emitted from the tip at an angle of about 90 degrees relative to a longitudinal axis of the tip.
6. An apparatus as recited in claim 5, wherein an interior of the caliper includes an angled surface at the tip that is configured to reflect the light at the angle.
7. An apparatus as recited in claim 6, wherein the angled surface is oriented at about 45 degrees relative to the longitudinal axis of the tip.
8. An apparatus as recited in claim 6, wherein the angled surface is configured to reflect the light at the angle using total internal reflection within the tip of the caliper.
9. An apparatus as recited in claim 1, wherein an end of the source line comprises a polished angled surface that is oriented at about 45 degrees relative to a longitudinal axis of the tip such that the light from the source line is emitted from the tip at an angle of about 90 degrees relative to the longitudinal axis of the tip.
10. An apparatus as recited in claim 1, wherein the handle and the caliper are integral.
11. An apparatus as recited in claim 1, wherein the first optical path includes: a first diffusion length where the light incident from the source line propagates in a straight direction within the sample; a path where the light deviates and is scattered from the target-volume of the sample; and a second diffusion length where the scattered light from the target-volume propagates in a straight direction before exiting the sample and into the detector line.

12. An apparatus as recited in claim 11, wherein the second optical path includes: a diffusion length where the light incident from the source line propagates in a straight direction within the sample; a path where the light backscatters to the non-target volume; a path where the light scatters from the non-target volume and enters the detector line.
13. The apparatus as recited in claim 1, wherein the sample is a tooth, the non-target volume is gum tissue and the target volume is a pulp region.
14. An apparatus comprising: a probe; a pair of fiber optic lines passing through an interior of the probe; wherein one of the fiber optic lines is a source line with a connector at an end of the source line; wherein another of the fiber optic lines is a detector line with a connector at an end of the detector line; wherein the probe includes a tip configured to engage opposite sides of a sample; wherein the source line is configured to receive light from an optical source having a coherence length; and wherein the tip is configured such that a difference between a first optical path along the source line and the detector line through a target volume of the sample and a second optical path along the source line and the detector line through a non-target volume of the sample exceeds the coherence length of the light to optically suppress signals from the non-target volume at the detector line such that the signals from the non-target volume do not interfere at the detector line and such that motion within the target volume is measured while motion within the non-target volume is suppressed.
15. The apparatus as recited in claim 14, wherein the probe comprises: a handle; a pivot; and a caliper including the tip.
16. The apparatus as recited in claim 15, wherein the pair of optical fibers pass through an interior of the handle, the pivot and the caliper.
17. The apparatus as recited in claim 14, wherein the source line is single mode and the detector line is multi mode.
18. The apparatus as recited in claim 14, wherein the probe is a tooth probe such that the tip is configured to engage opposite sides of a tooth of the sample, wherein the target volume is a pulp region within the tooth and wherein the scattered light from outside the target volume includes scattered light from gum tissue of the sample.
19. The apparatus as recited in claim 18, wherein the tip is further configured to push aside a gum tissue at a base of the tooth to access a cemento-enamel junction (CEJ) of the tooth.
20. The apparatus as recited in claim 18, wherein the coherence length of the light is in a range from about 2 mm to about 4 mm.
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